PROVISIONAL APPLICATION FOR PATENT

SYSTEM AND METHOD FOR DETERMINISTIC ARCHITECTURAL COMPILATION OF TRANSFORMER WEIGHTS FROM CORPUS STATISTICS

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June 17, 2026

Contents

1 TITLE OF THE INVENTION

SYSTEM AND METHOD FOR DETERMINISTIC ARCHITECTURAL COMPILATION OF TRANSFORMER WEIGHTS FROM CORPUS STATISTICS

2 CROSS-REFERENCE TO RELATED APPLICATIONS

The present application claims priority to and the benefit of any related research disclosures or internal developer records established within the legal framework governing provisional patent filings.

3 FIELD OF THE INVENTION

This invention relates generally to machine learning architectures, mathematical compiler design, and software engineering optimizations. More particularly, the present invention relates to systems, methods, and computer program products that take a corpus and an architecture as a functional hardware/software specification and directly compute highly optimized operational weight parameters (θ^*) for Transformer-based neural network models deterministically from corpus statistics and target architectural topologies, thereby entirely bypassing or minimizing traditional iterative backpropagation training trajectories.

4 BACKGROUND OF THE INVENTION AND PRIOR ART DESCRIPTION

State-of-the-art auto-regressive and bi-directional Transformer models (e.g., Large Language Models, Vision Transformers) are conventionally generated by initializing weight matrices with random or pseudo-random distributions and subsequently tuning those parameters across thousands or millions of iterations via stochastic optimization algorithms such as Stochastic Gradient Descent (SGD) or AdamW. This methodology treats the loss landscape of the neural network as an empirical black box, traversing it blindly via gradient approximations computed from mini-batches of data.

This classical paradigm suffers from severe technical and computational limitations:

1. **Extreme Resource Consumption:** Training medium-to-large-scale Transformer models requires millions of GPU/TPU hours, generating massive thermal and environmental overhead.

2. **Stochastic Convergence Risk:** Gradient descent paths are subject to chaotic variations, vanishing or exploding gradients, and high sensitivity to random initialization states (seeds).

3. **Theoretical Blind Spots:** Prior optimization techniques fail to exploit the structural predictability of the loss landscape. They treat the model's convergence path as a historical artifact of its specific training trajectory, rather than recognizing it as a deterministic consequence of the underlying interaction between static data distributions and architectural topology.

Consequently, a critical deficiency exists in the state of the art for a system that can bypass empirical trajectory optimization. There is an immediate requirement for an explicit "Transformer

Compiler" capable of mapping an input specification directly onto its optimal parameter configurations by computing the inherent geometric properties of the loss landscape analytically.

5 SUMMARY OF THE INVENTION

The present invention resolves the aforementioned inefficiencies by replacing stochastic gradient-based parameter discovery with direct mathematical weight synthesis.

The fundamental technical insight realized in this invention is that the geometry of a Transformer's loss landscape is fundamentally bound and fully determined by two static components: the global statistical properties of the input corpus and the geometric constraints of the target network architecture. Because this geometry is predictable prior to runtime execution, critical mathematical coordinates—such as the optimal basins of attraction and structural attention routing tables—can be isolated deterministically without relying on gradient trajectories.

The system is organized as a multi-phase compiler that transforms the raw input specification (`corpus.txt` and the target neural topology) through successive intermediate representations (IRs) to generate compiled weights (θ^*). The final component of the system acknowledges an irreducible mathematical boundary: the token embedding relaxation floor. Because embeddings are inextricably coupled to the language modeling prediction head, their optimal states depend on a joint fixed point of all variables. The compiler resolves this by isolating all parameters deterministically except the embedding layers, which are subsequently calibrated during an ultra-short, highly accelerated relaxation pass (e.g., approximately 25 Cross-Entropy optimization steps).

6 BRIEF DESCRIPTION OF THE DRAWINGS

A complete understanding of the present invention may be obtained by reference to the accompanying drawings, when considered in conjunction with the subsequent detailed description, in which:

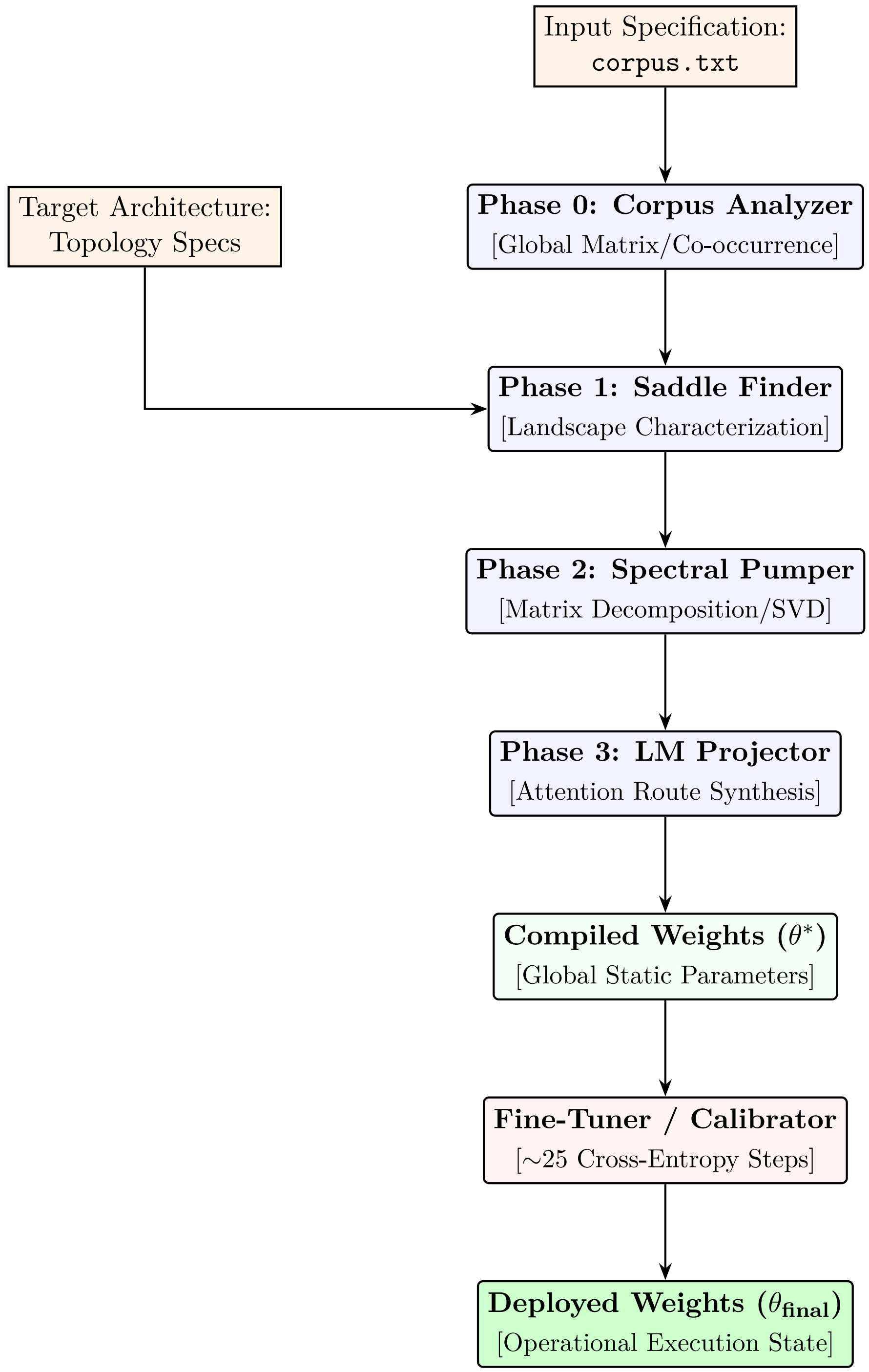

Figure 1: Block diagram illustrating the data flow, functional sub-components, and multi-phase execution sequence of the Transformer Compiler pipeline.

7 DETAILED DESCRIPTION OF THE INVENTION

The core operational logic of the Transformer Compiler relies on treating machine learning optimization as an explicitly deterministic compilation target. The system decouples weight generation into four discrete, composable mathematical phases.

7.1 Phase 0: Corpus Analyzer

The Corpus Analyzer serves as the ingest engine for the compiler. Rather than reading the text corpus sequentially or randomly sampling minibatches as in gradient descent, Phase 0 evaluates the textual corpus globally. It computes structural token dependencies, multi-order conditional probabilities, and creates a dense global co-occurrence matrix C_{ij} representing the statistical infrastructure of the dataset. Let V be the vocabulary size; the analyzer isolates the joint distributions across a context window W:

$$P(w_i, w_j) = \frac{\text{Count}(w_i, w_j \text{ within } W)}{\sum_{k,l} \text{Count}(w_k, w_l)} \tag{1}$$

7.2 Phase 1: Saddle Finder

The Saddle Finder maps the empirical statistical substrate generated in Phase 0 into the structural geometric bounds of the desired target architecture (e.g., layer depth L, dimension d_{model}, number of attention heads H).

Because the mathematical mechanics of standard Softmax attention interact with the token probabilities in predictable topologies, the loss function's analytical derivative zeroes can be predicted. Phase 1 calculates the coordinates of critical high-dimensional saddle points and isolates the primary stable basins of attraction prior to placing any weights into hardware memory. This eliminates the random search phase characterizing standard training.

7.3 Phase 2: Spectral Pumper

The Spectral Pumper initializes the structural transformations and projection matrices within the network. It takes the global co-occurrence mappings and computes high-dimensional matrix decompositions, focusing primarily on truncated Singular Value Decompositions (SVD) and singular value spectral density maps. For a targeted sub-layer projection matrix, the system extracts the dominant singular vectors:

$$C = U\Sigma V^T \implies W_{\text{proj}} \approx U_k \Sigma_k^{1/2} \tag{2}$$

This locks down the coordinate space of the residual streams and feed-forward sub-networks natively based on structural data density.

7.4 Phase 3: Language Model (LM) Projector

The LM Projector maps the analytical trajectories directly into the multi-head attention mechanisms (Queries, Keys, Values matrices) without executing any loss backward passes.

Using the basin structures mapped out in Phase 1 and the spectral dimensions from Phase 2, Phase 3 explicitly sets the attention heads to function as directional routers that match the structural transition matrices of the target language. The output produced by this stage is a globally synthesized, highly optimized set of compiled weights denoted as θ^*.

7.5 The Irreducible Computation Target: Embedding Relaxation

While the macro-level transformations, attention projections, and feed-forward transitions are deterministically structured from the global corpus properties, the token embedding matrix represents an irreducible computation limit. Because the input embedding coordinates are deeply tied to the output language modeling head, their combined optimal values depend on finding a joint fixed point across all dimensions simultaneously.

To resolve this without reverting to long training runs, the compiler injects the synthesized parameters θ^* directly into the model structure, anchoring the entire transformer architecture. The system then routes the model to a localized calibration unit. This unit performs an ultra-short embedding relaxation pass (restricted to approximately 25 total Cross-Entropy iterations over calibration token structures). This pass functions solely as a geometric adjustment phase that snaps the flexible embedding vectors into alignment with the hard-compiled global weight frameworks, yielding the operational, deployable parameter state θ_{final}

8 CLAIMS

We Claim:

1. A computer-implemented compiler system for synthesizing operational weights for a transformer-based neural network model, the system comprising:
   - an input interface configured to receive a design specification comprising a target textual corpus and a target neural architectural topology constraint;
   - a corpus analyzer configured to evaluate said target textual corpus to output a global matrix of token co-occurrence statistics;
   - a geometric saddle finder component configured to analytically isolate optimal basin structures of a high-dimensional loss landscape derived from the intersection of said global matrix of token co-occurrence statistics and said target neural architectural topology constraint;
   - an analytical weights synthesis pipeline comprising a spectral decomposition engine and an attention projection mapping engine, wherein said pipeline outputs a set of compiled global weights (θ^*) directly from said optimal basin structures without gradient descent calculations; and
   - an embedding relaxation optimization engine configured to perform a constrained sequence of cross-entropy calibration steps to align a token embedding matrix with said set of compiled global weights (θ^*) to generate a deployable operational model configuration (θ_{final}).

2. The system of claim 1, wherein said global matrix of token co-occurrence statistics maps higher-order conditional probability distributions across a predefined context token window width.

3. The system of claim 1, wherein said spectral decomposition engine relies on singular value decompositions of text feature matrices to establish static projection parameters for residual stream processing within said transformer-based neural network model.

4. The system of claim 1, wherein said attention projection mapping engine calculates Query-Key-Value projection matrix entries deterministically by matching attention routing vectors with token transition patterns derived by said corpus analyzer.

5. The system of claim 1, wherein said constrained sequence of cross-entropy calibration steps executed by said embedding relaxation optimization engine is bounded to an operational floor of fewer than one hundred training steps.

9 ABSTRACT OF THE DISCLOSURE

A system and method for a Transformer Compiler are disclosed which altogether eliminate or severely reduce classical iterative backpropagation trajectories by synthesizing neural network weights deterministically. The compiler ingests a design specification consisting of a target text corpus and an architectural topology design. A corpus analyzer processes the text corpus globally to establish token co-occurrence matrix structures. A geometric engine utilizes these matrices alongside the architectural constraints to map the critical saddle points and structural basins of attraction defining the network's loss landscape prior to weight initialization. Through spectral decomposition and language model projection mechanisms, optimized global weights (θ^*) are directly synthesized. An ultra-short, highly constrained embedding relaxation calibration step is subsequently applied to integrate joint parameter fixed points between token embeddings and the pre-computed global weights, outputting a fully functional, deployable model (θ_{final}).